

Embedded projects 2

Weekly exercises 8

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Mid-point review

Your task consists of three steps.

For this review you will present steps 1 and 2 **before** continuing to step 3.
(Deadline for steps 1 and 2 is [20.2.](#))

Step 1 is to plan and make a [flowchart](#) of how you think your car's code will work
(using PowerPoint, Paint, whatever tool you find online...)

Step 2 is to make [pseudocode](#) based on your flowchart

Step 3 is where you write your pseudocode into actual code and demonstrate your final product (and code) to teachers when it works according to the criteria
(Deadline for step 3 is 6.3.)

The criteria

- You drive your car with TURN and MOVE commands given from ESP/webpage
- Your car/lidar keeps track continuously that there are no walls/objects on your way
 - car must stop immediately if there are obstacles closer than 10 cm, in any case
 - if there are obstacles nearby (distance < 30 cm) the car moves slowly (~30 % of max speed)*
 - if there are no obstacles nearby (>30 cm) the car must move faster (~70 % of max speed)*
- Using ESP and 2-way communication, send info about lidar distance from Mega to ESP/webpage
 - if the lidar distance is less than 10 cm, display a “Warning!”-string on the webpage
- If webpage-user tries to drive forward when lidar distance is < 10 cm, your car must rotate 90 degrees at a time, **until** there are no obstacles closer than 10 cm and then drive the given distance.
 - turning must stop if there are walls all around the car (after finishing an entire 360° circle)